

SHYAM SUNDER

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EDUCATION

Alliance University, Bangalore 2023 – 2027 (Expected)
B.Tech, Computer Science and Engineering (AI & Machine Learning) CGPA: 8.45 / 10

EXPERIENCE

Software Engineer Intern – Enarin Business Solutions, Remote Mar 2026 – Jun 2026
Python, FastAPI, SQL, Redis, Docker, React, Git

- Built Python/FastAPI back-end services and React features for an AI-powered examination platform, covering document processing, automated evaluation workflows and authentication.
- Applied data structures and algorithms to automate multi-step evaluation workflows; wrote clean, documented Python and SQL.
- Implemented role-based access control with SQL and Redis for data persistence; containerised with Docker for reproducible deployments.
- Worked with senior engineers through code reviews, design discussions and debugging in Agile sprints.

Machine Learning Intern – Enarin Business Solutions, Remote Dec 2025 – Feb 2026
Python, NLP, LLMs, FastAPI

- Built Python/NLP data pipelines to clean, process and prepare candidate-job data for machine learning models, automating evaluation workflows.
- Integrated ML inference into FastAPI services, helping move AI features from prototype to production.
- Debugged data-processing workflows and collaborated with senior engineers on design and problem-solving.

Undergraduate AI Researcher – Alliance University, Bangalore Aug 2025 – Jan 2026
Python, TensorFlow/Keras, Scikit-learn, CNNs, Computer Vision

- Developed and evaluated CNN models for medical image (lung cancer) classification, reaching **98% accuracy** through systematic hyperparameter optimization; published in *Scientific Reports* (Nature Portfolio, Q1).

PROJECTS

Real-Time Campaign Optimization – Contextual Bandit (LinUCB) 2025
Python, NumPy, Pandas, Apache Kafka, PostgreSQL, Docker, Streamlit, SHAP

- Implemented a LinUCB contextual bandit from scratch in NumPy to select the best campaign channel per customer, using **21 engineered** RFM, behavioural and product-affinity features (Pandas, Parquet) on a synthetic dataset of 100K transactions and 10K customers.
- Streamed interactions through Kafka for online model updates with checkpointing and PostgreSQL logging; containerised with Docker.
- Built an A/B test simulator and Streamlit dashboards with SHAP explainability; the policy showed a **40% reward lift** over a random baseline (simulated data, $p < 0.001$).

PII Redaction System – Multimodal Document Redaction 2026
Python, PyTorch, YOLO, LayoutLMv3, TrOCR/OCR

- Built a multimodal PyTorch pipeline combining YOLO (region detection), LayoutLMv3 (layout understanding) and TrOCR/OCR (text extraction) to fuse visual and textual signals and redact PII in scanned documents.

CodeRAG – AI Code Intelligence Platform 2026
Python, FastAPI, Azure OpenAI, LangChain, FAISS, Neo4j, TypeScript, Docker

- Built a RAG-based VS Code extension on a FastAPI and Azure OpenAI service, using FAISS embeddings-based semantic search and AI agents for context-aware code retrieval across large codebases.
- Added a Neo4j graph layer modelling code structure (calls, imports, class hierarchies) and combined it with vector search into a hybrid retrieval pipeline; added structured logging, health checks and tests.

TECHNICAL SKILLS

Languages & Data: Python, SQL, Pandas, NumPy, JavaScript, Data Structures & Algorithms, OOP
Machine Learning: Scikit-learn, PyTorch, TensorFlow, Keras, CNNs, NLP, Computer Vision, Reinforcement Learning (Bandits), Feature Engineering, A/B Testing, Model Evaluation, Hyperparameter Optimization, SHAP
Data Engineering: Data Pipelines, PostgreSQL, Apache Kafka, Apache Spark, Redis, Neo4j
Cloud & Tools: Azure, Docker, Kubernetes, Git, GitHub Actions, CI/CD, FastAPI, Django, REST APIs, Streamlit, React
GenAI: LLM APIs (Azure OpenAI, Gemini), RAG, LangChain, FAISS, AI Agents, Prompt Engineering

ACHIEVEMENTS

- Published Researcher, Scientific Reports (Nature Portfolio, Q1):** CNN hyperparameter optimization for lung cancer classification.
- 3rd Place, Microsoft AI for Good Hackathon:** AI Resume and Skill Gap Analyzer.